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Git Repo : https://github.com/Ranjan-alok/cs3081-cloud-mini-project

1* What I Learnt :-

* Sequence-Level knowledge Distillation : I learned how to transfer capability from a large teacher model (phi-3.5) into a compact student model (Pythia-70m) by training the student directly on the teacher's generated text output rather than its internal weights.

* Why logit-Level Distillation : I understood the difference between hard-label Sequence distillation & soft-target logit distillation.

* Cloud-Native Deployment & Kubernets (K3d) : I learned how to containerize a FastAPI ~~intere~~ inference service & orchestrate it on a local k3d cluster using Deployments Traefik Ingress routing, readiness/liveness probes (/healthz & /readyz), CPU/memory resource limits, Horizontal Pod Autoscaling (HPA), And zero-downtime rolling updates (MaxUnavailable = 0).

* Cloud Network Security (VPC Subnetting & NACLs) : I Learned how to divide a /16 CIDR block into contiguous subnets ordered by host capacity (app, public, data, management) and configure stateless Network Access Control Lists (NACLs) where rule numbering order and explicit ephemeral return-port rules (1024-65535) are critical.

2* Challenges and Solutions :-

* Teacher Model Adding Conversational phase :-
Problem : During initial labeling runs, the phi-3.5 teacher model frequently added extra explanatory

text after the JSON block, causing raw JSON validity to drop to around 17%

Solution: I enforced JSON-constrained decoding in the ollama request payload (format: "json") with temperature set to 0 and a fixed seed (42), this eliminated empty or conversational replies & achieved 100% valid JSON across all 3000 commands.

*/ Local Resource Contraints & Cluster Execution :-

Problem: Running heavy container builds, local LLM labeling, and multi-replica kubernetes load testing simultaneously strained local hardware and virtualization limits

Solution: Alongside local modular testing, I utilized the Linux runner inside GitHub Actions to execute the full live-cluster K3d integration workflow, capturing clean, verifiable telemenry and load-test logs,

3. Observations & Key Findings. :-

* Distillation ~~Quanti~~ Quality :- The fine-tuned Student (V4) achived 99.0% JSON validity and 43.0% full frame accuracy capared to teacher's model 48.0% full frame accuracy (while the untrained base model score 0.0%). Training loss converged smoothly across the 3 epochs (0.257 → 0.105 → 0.072), while the 70M parameters students successfully learned the JSON schema & intent classification (92.0)% of its small capacity slightly underfits complex slot extraction rules.

* Data cleaning Metrics : Out of 3000 raw commands labeled by the teacher, 0 failed JSON parsing 1 had wrong keys, 3 had invalid intents, 86 contained hallucinated slot values not present in the source text and 49 were duplicates - leaving 2861 clean training records verified by deterministic SHA-256 checksum.

4. Future Improvements :

* Model Capacity & Request Batching : Training for more epochs, refining the teacher prompt to reduce the 86 false-hallucination rejections, or using a slightly larger Student model (160M parameters) would improve full-frame accuracy. Adding dynamic request batching in FastAPI would significantly improve CPU inference throughput and lower p95 latency.

* Dynamic work Queue for labeling : Static modulo sharding cannot rebalance work if one worker pod runs slower than another replacing it with a shared message Queue (such as Redis or RabbitMQ) would eliminate stragglers.